Definition of SNMP Trap

An **SNMP trap** is an asynchronous notification (or alert) sent by an SNMP-enabled device (such as a router, switch, or server) to an SNMP manager to inform it about significant events or status changes in real-time. Unlike traditional SNMP polling, where the SNMP manager requests information, traps are **sent automatically** when certain conditions are met.

How SNMP Traps Work

1. **A network device detects an event** (e.g., an interface goes down, high CPU usage, or unauthorized login).
2. **The SNMP agent on the device sends a trap message** to the SNMP manager (monitoring system).
3. **The SNMP manager receives and logs the trap**, potentially triggering alerts, emails, or automated responses.

Examples of SNMP Traps

Here are some common SNMP traps with explanations:

1. Link Down (`linkDown` OID: 1.3.6.1.6.3.1.1.5.3)

- Sent when a network interface goes **down** (disconnected or failed).
- Example scenario:
  - A switch port connected to a server loses connection.
  - The switch sends a **linkDown** trap to the SNMP manager.
  - The SNMP manager generates an alert: **"Critical: Port 1 on Switch X is down!"**

2. Link Up (`linkUp` OID: 1.3.6.1.6.3.1.1.5.4)

- Sent when a previously down interface **comes back up**.
- Example scenario:
  - A network cable is reconnected to a router.
  - The router sends a **linkUp** trap.
  - The SNMP manager logs: **"Info: Interface G0/1 on Router A is back up."**

3. Authentication Failure (`authenticationFailure` OID: 1.3.6.1.6.3.1.1.5.5)

- Sent when an unauthorized SNMP request is detected.
- Example scenario:
  - Someone tries to access a network switch using the wrong SNMP community string.
  - The switch sends an **authenticationFailure** trap.

- The SNMP manager generates a security alert: **"Warning: Unauthorized SNMP access attempt detected!"**

4. CPU High Usage (`cpuHigh` - Vendor-Specific OID)

- Sent when CPU utilization exceeds a defined threshold.
- Example scenario:
  - A firewall's CPU usage reaches **90%** due to high traffic.
  - The firewall sends a **cpuHigh** trap.
  - The SNMP manager alerts: **"Critical: CPU usage on Firewall Y exceeded 90%!"**

5. Configuration Change (`configChange` - Vendor-Specific OID)

- Sent when a device configuration is modified.
- Example scenario:
  - An admin makes changes to a router's **ACL settings**.
  - The router sends a **configChange** trap.
  - The SNMP manager logs: **"Info: Configuration changed on Router Z."**

Conclusion

SNMP traps are **critical for real-time network monitoring** and help administrators detect and respond to issues quickly without continuously polling devices.

The given SNMP (Simple Network Management Protocol) configuration commands are used to enable and configure SNMP traps on a network device (such as a router or switch). Let's break them down:

---

1. **`snmp-server host [log host] [version] [community string] [trap-type]`**

This command is used to specify the destination (host) where SNMP traps should be sent.

Breakdown of Parameters:

- **`snmp-server host`** → Configures the SNMP trap receiver (destination).
- **`[log host]`** → The IP address or hostname of the SNMP manager (monitoring system) that will receive the traps.
- **`[version]`** → Specifies the SNMP version used (v1, v2c, or v3).
- **`[community string]`** → This is like a password used for authentication between the SNMP agent and manager (for SNMP v1/v2c).

- **`[trap-type]`** → Defines which types of SNMP traps should be sent (e.g., linkdown, linkup, authentication, CPU threshold alerts, etc.).

Example Usage:

```
snmp-server host 192.168.1.100 version 2c public linkdown
```

- Sends SNMP traps to the SNMP manager at `192.168.1.100`
- Uses **SNMP v2c**
- The **community string** is `public`
- The **trap-type** is `linkdown` (sends a trap when an interface goes down)

---

2. `snmp-server enable traps [trap-type]`

This command globally enables SNMP traps for specific events.

Breakdown of Parameters:

- **`snmp-server enable traps`** → Enables SNMP trap notifications.
- **`[trap-type]`** → Specifies which types of traps should be enabled (e.g., `config`, `syslog`, `cpu`, `link-status`, etc.).

Example Usage:

```
snmp-server enable traps config
```

- Enables SNMP traps for configuration changes.

```
plaintext
CopyEdit
snmp-server enable traps link-status
```

- Enables SNMP traps for **interface status changes** (up/down).

How These Commands Work Together

1. **Step 1:** `snmp-server enable traps [trap-type]` → This enables the specific SNMP trap feature.
2. **Step 2:** `snmp-server host [log host] [version] [community string] [trap-type]` → This configures where the traps should be sent.

Complete Example

```
snmp-server enable traps link-status
snmp-server host 192.168.1.200 version 2c public linkdown linkup
```

- Enables SNMP traps for **link status changes**.
- Sends **linkdown** and **linkup** traps to `192.168.1.200` using SNMP v2c with the `public` community string.

Conclusion

- **`snmp-server enable traps`** turns on trap notifications for a specific event type.
- **`snmp-server host`** defines **where** the traps should be sent and with what SNMP version and authentication.
- Both commands must be used together to properly send SNMP traps to a monitoring system.

Let me know if you need further clarifications!